

Quick Start Guide: AMD MicroBlaze V Soft Processor for Vitis Unified 2024.1

INTRODUCTION

This Quick Start Guide will walk you through creating a basic AMD MicroBlaze™ V processor system using processor preset designs. Additional resources and information can be found on the following page to help you tailor a MicroBlaze V processor system to your exact design specifications. Features include:

- Royalty Free
- Highly Configurable
- Open Standard ISA
- More Configurable Extensions
- Compressed Instructions
- Free Development Tools

What is the MicroBlaze V Processor?

MicroBlaze V is a soft processor core optimized for embedded applications on AMD FPGA and adaptive SoCs supported by the AMD Vivado™ Design Suite. The MicroBlaze V processor is easy to use and delivers the flexibility to select the combination of peripherals, memory, and interfaces as needed.

The MicroBlaze V processor can be configured with ease. By selecting features like RISC-V Instruction Set Architecture options, pipeline, and cache sizes, you can build the most efficient processor for your application.



MicroBlaze V can be used as a stand-alone processor or as a co-processor in other AMD FPGA and adaptive SoC systems. It can also be configured to add tamper protection and fault protection by configuring in lock-step mode as well as providing single-event upset mitigation with Triple Modular Redundancy. Designs with one or more processors can be debugged simultaneously using the AMD Vitis™ Unified Software Platform.

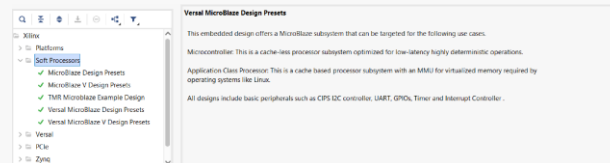


BEFORE YOU BEGIN

This Quick Start Guide assumes you are targeting an AMD development board. If this board comes from an AMD board partner, you need to download the latest boards and example projects within the Vivado tools. See the FAQ (next page) for links to some of our partners.

HARDWARE DEVELOPMENT

- Start Vivado Design Suite (2024.1 or later).
- Under **Tools** select **Vivado Store**. Select the Boards tab then click **Refresh** to download the latest version of the catalog.
- When download is complete, choose **Open Example Project**.
- When Wizard opens, read information text and click **Next**.
- Before choosing a template, click **Refresh again** at the bottom left.
- From the templates, select **MicroBlaze V Design Presets** or **Versal™ MicroBlaze V Design Presets**, then click **Next**.

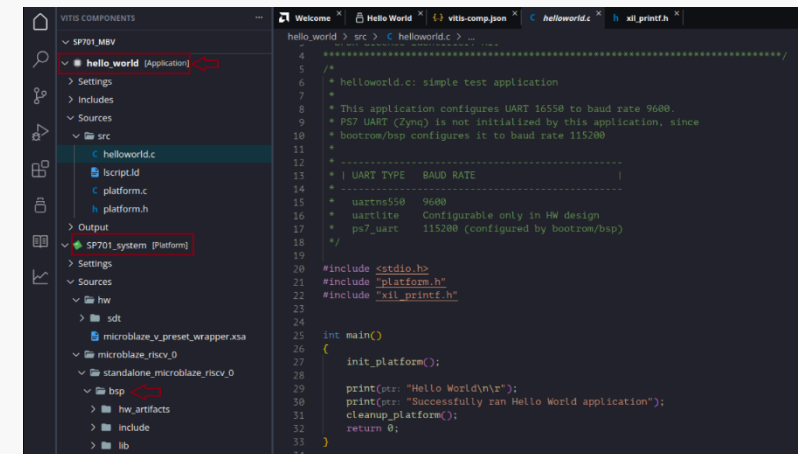


Enter a project name and location for project files and click **Next**.

- Choose the target board and click **Next**.
- Choose **Microcontroller** and click **Next**.
- Now click **Finish** to create the project and the Block Design will open.
- Double click the **MicroBlaze V** block in the diagram.
- Under Predefined Configurations, notice that there are different configuration settings. Click **Cancel** to keep the current settings.
- To now save the design, press **Ctrl + S** or click **File→Save Block Design**.
- Next, to generate the bitstream, which contains configuration data for the device, choose **Generate Bitstream** or **Generate Device Image**.
- Launch Synthesis and Implementation runs, click **Yes**. Build status is shown in the top right corner of Vivado. *Ready* indicates completion.
- When completed, click **OK** to Open Implemented Design.
- From the main toolbar, click **File** and select **Export→Export Hardware**. Check the box to **Include Bitstream** or **Include Device Image** and make sure it is exported to the same project location and click **OK**.
- To start software development with the MicroBlaze V processor, select **Tools → Launch Vitis IDE** from the main toolbar. Vitis will now open and import the hardware platform, including the MicroBlaze V.

SOFTWARE DEVELOPMENT

- When the Vitis tools launch, click **Open Workspace** and create a new workspace folder.
- Click **Create Platform Component**, and then name and place the component location within your workspace.
- Choose your XSA file by clicking **Browse**.
- After your Platform component has been generated, click **Examples** from the Welcome Menu and click **Hello World**.
- Click on **Create Application Component from Template**.
- Set project name to **Hello_World** with no spaces.
- Verify your project location, and select your **Platform Component** and click **Next**.
- Expand the **src** folder and double click **helloworld.c** to view and edit the source code.
- Click on the build  button to build your project.
- You will see two important folders in the **Explorer** window: **Hello_world** contains all binaries, .C, and .H (Header) files **microlaze_v_preset_wrapper** includes the board support package (bsp) folder - Software drivers, software specification, and Makefile.



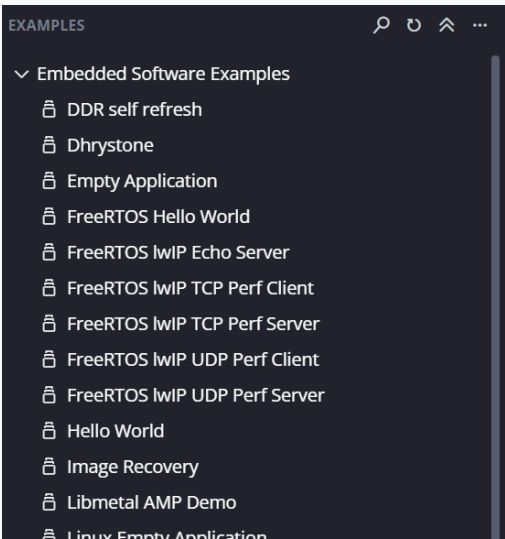
- Make sure your target board is turned on and connected to the host PC via the USB-JTAG port; this port also serves as the USB-UART connection to the MicroBlaze V processor.
- On the top toolbar, click **Vitis → Program Device** then **Program** again to program your device with your hardware design.

SOFTWARE DEVELOPMENT (cont'd)

- Start a serial terminal program of your choice.
- Download a serial terminal program of your choice.
- Select your serial port and set your Baud Rate to 115200.
- Now go to the debug page by clicking on the **Debug** tab  on the left.
- Program your board by hitting **Start Debug.** 
- Now run the application by clicking **Continue.** 
- Your program will run, and you should see “Hello World” pop up inside of your Serial Terminal.

```
Opened with baud rate: 115200
Hello World
Successfully ran Hello World application
```

- Congratulations! You have created your first MicroBlaze V processor application.
- Now you can try building and running other example applications, like the ones provided:



(FreeRTOS is not supported and will be removed in 2024.2)

FAQs and ADDITIONAL RESOURCES

- **How do I load third-party boards into a Vivado example designs?**
See [UG994](#).
- **Where can I find specific details about the MicroBlaze V processor?**
Go to: [UG1629](#)
- **How can I estimate the size and performance for my configured MicroBlaze V processor?**
Read [UG1629](#) under the Performance and Utilization section.
- **Where do I go for more details about creating a Vivado tool design?**
Visit our [Vivado Design Hubs](#).
- **Do I need to be in the Vivado tools to start Vitis?**
No. Vitis is a unified software platform that can be launched independently from Vivado. However, you'll need a Vitis platform or create a new platform from the hardware (.xsa) file to target for software development.
- **What do I do if the board I'm targeting is not on the list?**
Many board vendors provide board files and platforms that can be added to Vivado and Vitis. Contact the specific manufacturer for these files.
- **What if I need to make changes to my hardware design?**
Close Vitis and make the required HW design edits in the Vivado tools, then follow the sequence for bit file generation. This updated hardware design must then be exported from the Vivado tools and imported into Vitis as a new platform.
- **How do I expand the capability of my evaluation board?**
[Pmods](#), [Arduino shields](#), [Click boards](#), and [FMC cards](#) can be used to expand the capabilities of our evaluation boards.
- **How do I create a bootable flash image that contains my device image and application?**
See [UG898](#):
 - In Vivado, **Tools** → **Associate ELF Files...**
 - In Vitis, **AMD** → **Program FPGA** (select ELF for MicroBlaze V).

- **What happens when I export hardware and launch Vitis Unified?**
AMD Support Archive (.xsa) file is created. This file contains HW specifications, IP interfaces, external signal information, and local memory address information. This is used by Vitis to create a hardware platform.
- **How do I communicate between Zynq 7000 SoC and MicroBlaze?**
See this QTV on YouTube: [Zynq and MicroBlaze IOP Block, OCM, and Memory Resource Sharing](#). (This video was developed for MicroBlaze but is still applicable to MicroBlaze V.)
- **How much FPGA memory can the MicroBlaze V processor access?**
MicroBlaze V systems can be created that access all the available memory on an FPGA. But this comes at the cost of lower F_{MAX}. Typical MicroBlaze V implementations use 128KB or less.
- **What libraries are supported in Vitis for MicroBlaze V?**
See [UG643 - OS and Libraries Guide](#).
- **Can I run an RTOS on the MicroBlaze V processor?**
Yes. Any RTOS that does not need memory protection or memory management can be run on MicroBlaze V. (Memory protection and memory management will be added to MicroBlaze V in the future.)
- **Are we able to add custom extensions?**
No, the core is closed source, and we don't currently have a plan to make it open source or add an API for custom instructions.

Resources

- [MicroBlaze V Soft Processor Core Product Page](#)
- [Document Navigator Embedded Hub](#)
- [Vivado Design Suite Tutorials](#)
- [Vitis™ Unified Software Platform Documentation](#)
- [Knowledge Base Answer Records](#)
- Third-Party Partner Boards
[Avnet](#) | [Digilent](#) | [Trenz](#) | [Enclustra](#) | [iWave](#) | [MYiR](#) | [ALINX](#)